

Mr. ABHISHEK KUMAR PANDEY

PERSONAL DATA

NAME: Abhishek Kumar Pandey
GENDER: Male
PLACE AND DATE OF BIRTH: Balrampur, India | 26 July 1999
ADDRESS: Terhi Bazar Near G.I.C. Balrampur, Uttar Pradesh, India-271201
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WORK EXPERIENCE

since 07/2022	Ph.D. in CSE from the Center for Security, Theory and Algorithmic Research (C-STAR), under the supervision of Prof. Ashok Kumar Das , International Institute of Information Technology, Hyderabad.
09/2020-08/2021	Researcher at Information Technology Laboratory, Babasaheb Bhimrao Ambedkar University, (A Central University), Lucknow under the supervision of Prof. R. A. Khan.

SCIENTIFIC EDUCATION

Proposal Defense: 09-07-2026	Thesis Proposal Defended. Title: <i>Design and Analysis of Secure Machine Learning-Driven Authentication Mechanisms for Medical Cyber-Physical Systems</i>
07/2022 - Present Laboratory	RESEARCH SCHOLAR. International Institute of Information Technology, Hyderabad. Supervisor: Prof. Ashok Kumar Das Centre for Security, Theory and Algorithmic Research (C-STAR)
07/2018 - 07-2020 Department	M.SC. CYBER SECURITY Babasaheb Bhimrao Ambedkar University (A Central University), Lucknow Supervisor: Prof. R. A. Khan Department of Information Technology
07/2015 - 06-2018 Department	BACHELOR OF COMPUTER APPLICATION (BCA) Siddhartha University, Kapilvastu, India "University Gold Medalist" Department of Computer Application

RESEARCH INTEREST

- **Broad:** Understanding and classifying various aspects of Cyber Security/Information Security and its different factors for advanced authentication.

- **Current Focus:** Secure Authentication, AI/ML Security, Cyber Physical System, Data Security, Health Informatics, Malware Analysis.

VISITING RESEARCHER

Under UTFORSK Project “Cyber Security for Critical Infrastructure”

17-Nov-2025 - 22-Nov-2025	EXCHANGE PROGRAM RESEARCHER University of South-Eastern Norway (USN) Hasbergs vei 36, 3616 Kongsberg, Norway Host: Prof. Mohsen Toorani
Department	Department of Science and Industry Systems
23-Nov-2025 - 29-Nov-2025	EXCHANGE PROGRAM RESEARCHER NTNU – Norwegian University of Science and Technology Teknologivegen 22, 2815 Gjøvik, Norway Host: Prof. Sokratis Katsikas
Department	Department of Information Security and Communication Technology

PUBLICATIONS [DETAILED]

BOOK CHAPTERS [CH-1 : CH-3]

- CH-3 [Abhishek Kumar Pandey](#), Ashutosh Kumar Tripathi, Alka Agrawal, Rajeev Kumar, Raees Ahmad Khan. **Selecting and Assessing the Importance of Malware Analysis Methods for Web-based Biomedical Services through Fuzzy based Decision Making Procedure.** *Recent advances in Nature Inspired Optimization Algorithms, Techniques and Biomedical Applications.* De-Gruyter, 2020.
- CH-2 [Abhishek Kumar Pandey](#), Ashutosh Tripathi, Gayatri Kapil, Virendra Singh, Mohd. Waris Khan, Alka Agrawal, Rajeev Kumar, Raees Ahmad Khan. **Trends in Malware Attacks-Identification and Mitigation Strategies.** *Critical Concepts, Standards, and Techniques in Cyber Forensics.* IGI Global, 2019.
- CH-1 [Abhishek Kumar Pandey](#), Ashutosh Tripathi, Gayatri Kapil, Virendra Singh, Mohd. Waris Khan, Alka Agrawal, Rajeev Kumar, Raees Ahmad Khan. **Current Challenges of Digital Forensics in Cyber Security.** *Critical Concepts, Standards, and Techniques in Cyber Forensics.* IGI Global, 2019.

JOURNAL ARTICLES [J-1 : J-14]

- J-14 Snehal Jain, [Abhishek Kumar Pandey](#), Ashok Kumar Das, Shantanu Pal, and Youngho Park. **Post-Quantum Secure Lattice-Based Authentication Scheme for IoT-Enabled Crop Recommender System.** *IEEE Internet of Things Journal*, June 2026. (2025 SCI Impact Factor: 8.7)
- J-13 [Abhishek Kumar Pandey](#), Ashok Kumar Das, Shantanu Pal, Bharat K. Bhargava, and

Chengzu Dong. **Authenticated Key Agreement Scheme for Securing IoMT-Based Consumer Smart Healthcare Using Reinforcement Learning-as-a-Service.** *IEEE Transactions on Consumer Electronics*, June 2026. (2025 SCI Impact Factor: 9.9)

- J-12 Rishita Verma, Gaurav Indra, [Abhishek Kumar Pandey](#), Ashok Kumar Das, and Vivekananda Bhat K. **ZT-RIASE: Zero Trust-Resilient Identity Attestation for Securing Smart Industrial IoT Environment.** *Scientific Reports (Nature)*, May 2026. (2024 SCI Impact Factor: 3.9)
- J-11 [Abhishek Kumar Pandey](#), Basudeb Bera, Anusha Vangala, Ashok Kumar Das, Youngho Park, and Biplob Sikdar. **Quantum-Safe Digital Twin Authentication for ML-Driven Early Disease Detection in Healthcare Systems.** *IEEE Journal of Biomedical and Health Informatics*, February 2026, DOI: 10.1109/JBHI.2026.3667781. (2024 SCI Impact Factor: 6.8)
- J-10 [Abhishek Kumar Pandey](#), Vincenzo Sammartino, Anusha Vangala, Ashok Kumar Das. **Blockchain-Enabled Quantum-Resistant Hybrid Isogeny-Based Signcryption Framework for Digital Twin-Assisted Healthcare Systems.** *IEEE Communications Standards Magazine*, February 2026, DOI: 10.1109/MCOMSTD.2026.3665935.
- J-9 [Abhishek Kumar Pandey](#), Sherali Zeadally, Ashok Kumar Das, Neeraj Kumar. **Secure Integration of Cyber Engineering and Medical Cyber-Physical System: A Survey and Open Issues.** *ACM Computing Surveys*, Vol. 58, No. 1, Article No. 9, pp. 1-36, January 2026, DOI: 10.1145/3757731. (2024 SCI Impact Factor: 28)
- J-8 Rohan Tripathi, Ritam Rao, Vivekananda Bhat K, [Abhishek Kumar Pandey](#), Ashok Kumar Das. **Secure Blockchain Integration Approach for Knowledge Discovery in Industrial Internet of Things.** *IEEE Open Journal of the Communications Society*, Vol. 6, pp. 4772-4787, 2025, DOI: 10.1109/OJCOMS.2025.3574816. (2024 SCI Impact Factor: 6.1)
- J-7 [Abhishek Kumar Pandey](#), Ashok Kumar Das, Mohammad Wazid, Kuljeet Kaur, Youngho Park, and Mohammad Mehedi Hassan. **Uncrewed Aerial Vehicles Empowering Secure Authentication in Cognitive IoMT for Transformative Knowledge Discovery in Data.** *IEEE Internet of Things Journal*, Vol. 12, No. 9, pp. 11329-11346, May 2025, DOI: 10.1109/JIOT.2025.3549720. (2024 SCI Impact Factor: 8.9)
- J-6 [Abhishek Kumar Pandey](#), Ashok Kumar Das, Rajeev Kumar, Joel J. P. C. Rodrigues. **Secure Cyber Engineering for IoT-Enabled Smart Healthcare System.** *IEEE Internet of Things Magazine*, Vol. 7, No. 2, pp. 70-77, March 2024, DOI: 10.1109/IOTM.001.2300172.
- J-5 Mohammad Zarour, Mamdouh Alenezi, Md Tarique Jamal Ansari, [Abhishek Kumar Pandey](#), Masood Ahmad, Alka Agrawal, Rajeev Kumar, Raees Ahmad Khan. **Ensuring Data Integrity of Healthcare Information in the Era of Digital Health.** *Healthcare Technology Letters*, 2021.
- J-4 Mamdouh Alenezi, [Abhishek Kumar Pandey](#), Richa Verma, Mohd Faizan, Shalini Chandra, Alka Agrawal, Rajeev Kumar, Raees Ahmad Khan. **Evaluating the Impact of Software Security Tactics: A Design Perspective.** *CMC-Computers, Materials & Continua.*, 2020.

- J-3 Alka Agrawal, [Abhishek Kumar Pandey](#), Abdullah Baz, Hosam Alhakami, Wajdi Alhakami, Rajeev Kumar, Raees Ahmad Khan. **Evaluating the Security Impact of Healthcare Web Applications through Fuzzy based Hybrid Approach of Multi-Criteria Decision-Making Analysis.** *IEEE Access. Volume 8,* 2020.
- J-2 Rajeev Kumar, [Abhishek Kumar Pandey](#), Abdullah Baz, Hosam Alhakami, Wajdi Alhakami, Alka Agrawal, Raees Ahmad Khan. **Fuzzy-Based Symmetrical Multi-Criteria Decision-Making Procedure for Evaluating Impact of Harmful Factors of Healthcare Information Security.** *Symmetry. Volume 12, pp. 664,* 2020.
- J-1 [Abhishek Kumar Pandey](#), Asif Irshad Khan, Yoosef B. Abushark, Md Mottahir Alam, Alka Agrawal, Rajeev Kumar, Raees Ahmad Khan. **Key Issues in Healthcare Data Integrity: Analysis and Recommendations.** *IEEE Access. Volume 8, Issue 3, pp. 40612-40628,* 2020.

INTERNATIONAL CONFERENCE PAPERS [C-1 : C-4]

- C-4 [Abhishek Kumar Pandey](#), Debnath Ghosh, Mohammad S. Obaidat, Ashok Kumar Das, Shantanu Pal, Ravi Kumar Kappagantu, Srinivas V Katakam, and Jitendra Chougala. "Quantum-Resistant Security Protocol for Digital Identity Verification Using KYC-as-a-Service in 6G-Enabled Banking Applications," in **IEEE International Conference on Computer, Information and Telecommunication Systems (CITS 2026)**, 22-24 July 2026, Piraeus-Athens, Greece.
- C-3 Debnath Ghosh, [Abhishek Kumar Pandey](#), Prithwi Bagchi, Ashok Kumar Das, Shantanu Pal, Ravi Kumar Kappagantu, Srinivas V Katakam, and Jitendra Chougala. "Securing Financial Transactions Using DLT-Enabled Quantum-Safe Authentication Scheme," in **22nd IEEE International Wireless Communications and Mobile Computing (IWCMC'26)**, 1-6 June 2026, Shanghai, China. **[Rank: B]**
- C-2 [Abhishek Kumar Pandey](#), Prithwi Bagchi, Debnath Ghosh, Ashok Kumar Das, Ravi Kumar Kappagantu, Srinivas V Katakam, and Jitendra Chougala. "Post-Quantum Secure Lattice-Based Authentication Scheme for IoT-Enabled Contactless Smart Payments," in **7th IEEE Computers, Communications and IT Applications Conference (ComComAp 2025)**, 14-17 December 2025, Madrid, Spain.
- C-1 [Abhishek Kumar Pandey](#), Debnath Ghosh, Prithwi Bagchi, Ashok Kumar Das, Ravi Kumar Kappagantu, Srinivas V Katakam, and Jitendra Chougala. "Post-Quantum DLT Framework for Secure Financial Transactions in Banking Applications," in **18th IEEE International Conference on COMMunication Systems & NETWORKS (COMSNETS 2026)**, Cybersecurity and Privacy (CSP) Workshop, January 6 - 10, 2026, Bangalore, India. **[Best Paper Award]**

COMMITTEE MEMBER (CONFERENCES/WORKSHOP)

- Technical Program Committee (TPC) Member for **7th Computers, Communications and IT Applications Conference (ComComAp 2025)** at Madrid, Spain from 14-17 December 2025.
- Organizing Committee Member for **5-Day National Workshop and Boot-camp on Emerging Trend in Post-Quantum Security (NWPQSec 2026)** at IIIT Hyderabad from 6-10 March 2026.

- Organizing Committee Member for **5-Day National Workshop and Boot-camp on Cyber Security and Blockchain (NWBCSB 2025)** at **IIIT Hyderabad** from 1-5 March 2025.

PROFESSIONAL RESPONSIBILITIES

- Reviewer for IEEE Transactions on Information Forensics and Security (TIFS).
- Reviewer for Scientific Reports (Nature).
- Reviewer for IEEE Open Journal of the Communications Society.
- Reviewer for KSII Transactions on Internet and Information Systems.
- Reviewer for IET Communications.
- Reviewer for IEEE Access.
- Reviewer for Security Journal, Palgrave, Springer Nature.
- Reviewer for BMC Medical Informatics and Decision Making, Springer Nature.
- Reviewer for BMC Primary Care, Springer Nature.

SCIENTIFIC SOCIETY MEMBERSHIPS

- **Professional ACM Member (Member Number: 7388884)**
- A Member of International Computer Science and Engineering Society (ICSES) (Membership ID: 9243).
- A Member of Internet Society India – Global Member (Membership ID: 2221065).
- A Member of International Association of Engineers (IAENG) (Membership ID: 267366).

Abhishek Kumar Pandey
PhD Scholar, IIIT Hyderabad



August 17, 2026
